OUMALKAIRE ABOUBAKAR ABDI

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SUMMARY

Computer Science graduate (GPA 3.7/4.0) with software engineering and machine learning experience. Builds reproducible ML and data pipelines across audio signal processing, computer vision, geospatial biodiversity data and real-time systems, and shipped a full-stack client product as an AI Software Engineer Intern. Skilled in Python, R, SQL, TypeScript and Rust.

EDUCATION

Asia Pacific University of Technology & Innovation (APU) Kuala Lumpur, Malaysia
B.Sc. (Hons) Computer Science, dual award with De Montfort University, Leicester (UK) Expected Nov 2026

- **GPA:** 3.7/4.0 | Vice-Chancellor's Award for Academic Excellence, 2023/2024

Lycée Français de Koweït Kuwait City, Kuwait
French Baccalaureate, specialization in Mathematics and Computer Science 2023

- Graduated with highest honors, ranked 2nd of 80; Computer Science 19/20, Mathematics 18/20.

EXPERIENCE

AI Software Engineer Intern, Keenify Sep 2025 – Jan 2026

- Built a Christmas gift-box recommendation product end to end in 2 weeks with Python/FastAPI and TypeScript, generating combinations from a 43-product catalog with PDF report export; approved by the internship supervisor and 2 clients and launched on the client-facing website.
- Diagnosed a data-integrity bug in a production dashboard that was wrongly flagging users, traced it to mismatched source and stored data, and documented the root cause and fix.
- Built 2 n8n automation workflows with webhooks, including one that turns Fathom meeting transcripts into shared-drive action-item documents with owners.
- Wrote a 50-page proposal for SCCCI on applying AI to its operational challenges.
- Made 3 web apps mobile-responsive and delivered client-requested features and fixes in TypeScript, including a print-ready calendar, job-posting management and a database schema fix for employee offboarding.

Teaching Assistant, System and Network Administration, Asia Pacific University Jan 2026 – Present

- Taught 400+ students across 3 batches of the module, with 7 lab classes of 30+ students and 14 hands-on lab practicals per batch.
- Guide students through Linux networking, file sharing and system administration tasks, troubleshooting configuration issues live in class.
- Mentor and advise students on their hackathon projects.

PROJECTS

AutoGuard AI: Acoustic Fault Detection and LLM Diagnostic System for Equitable Repair 2026
Final Year Project, APU / De Montfort University | Supervisor: Ts. Dr. Manimegalai A/P Rajenderan

- Built to reduce the "pink tax" in car repair: research shows women who appear uninformed are quoted ~$20 more for the same repair, and the gap disappears once they know the market price (Busse et al., 2017).
- Surveyed 30+ drivers (52% women): 96.7% had felt overcharged by a mechanic and 76.7% found repair explanations confusing.
- Trained a CNN in TensorFlow/Keras on Mel-spectrograms to classify 6 fault types from audio, reaching 93% test accuracy on a public benchmark dataset.
- Deployed the model offline in a React Native app using TensorFlow Lite and INT8 quantization (12x smaller), flagging uncertain readings instead of guessing.
- Built a RAG chatbot with Gemini, pgvector (Supabase PostgreSQL) and FastAPI that explains diagnoses in 6 languages, plus a fair-price estimator using World Bank PPP data so users know the market price before they visit a mechanic.

Wikipedia Real-Time Monitor: Real-Time Data Pipeline in Rust 2026

- Built a Rust pipeline (Tokio, Axum) that ingests the live Wikipedia edit stream under a 2 ms per-event deadline, with a React dashboard for latency and throughput.
- Benchmarked async vs. threaded concurrency with Criterion: async cut p99 tail latency 2.3x (1.47 ms vs. 3.38 ms) at equal throughput of ~285K events/s.
- Added priority scheduling, backpressure and a watchdog with automatic degraded mode, with zero dropped packets in normal live operation.

RedSeaReefGap: Geospatial Sampling-Gap Analysis of Red Sea Reef-Fish Records 2026

- Built an R pipeline that pulls GBIF Occurrence API records for 6 reef-fish families in the Red Sea and quantifies spatial and temporal sampling bias.
- Found that 70–85% of records per family come from a single country, with 70–82% dating from 2015 onward.

- Automated the full analysis with GitHub Actions CI and validation checks, publishing results to a live GitHub Pages site on every push.

AquaSize AI: Reef-Fish Detection with Computer Vision 2026

- Trained a YOLOv11 object-detection model in Python on a 680-image dataset spanning 13 reef-fish families, reaching 76% top-1 accuracy on a held-out test set.
- Diagnosed undertraining by comparing 100- and 300-epoch runs, and evaluated with per-class recall and confusion matrices.
- Stress-tested accuracy under 8 simulated catch-photo conditions with Albumentations and deployed a live web demo.

SKILLS

Programming: Python, R, SQL, TypeScript, JavaScript, Rust, Bash

Machine Learning & AI: TensorFlow/Keras, TensorFlow Lite, scikit-learn, CNNs, YOLO, computer vision, model quantization, RAG, embeddings, prompt engineering, LLM APIs (Gemini, Groq)

Data & Signal Processing: NumPy, SciPy, librosa, Matplotlib, ggplot2, FFT, Mel-spectrograms, data augmentation, model evaluation

Software & Tools: FastAPI, REST APIs, PostgreSQL, pgvector, Supabase, React, React Native, Tokio, Axum, Git/GitHub, GitHub Actions CI/CD, Linux, n8n, Render, Figma

Research Methods: Survey design, quantitative analysis, sampling-bias analysis, geospatial analysis, literature review, reproducible pipelines

Professional: Agile/Scrum, stakeholder communication, technical writing, project planning, teaching and workshop delivery

LEADERSHIP & ACTIVITIES

Professional Development Director (Board Member), Rotaract Club of APU Jun 2024 – Jun 2025

- Organized and delivered a hands-on web-development workshop for 60+ students, each building and publishing a portfolio website.

Student Ambassador, Asia Pacific University & ReSkills Apr 2025 – Jan 2026

- Represented the university at outreach events for prospective students and families.

JomHack Hackathon

- Designed a multi-agent architecture to unify fragmented enterprise data silos, built under a no-AI-generated-code rule.

CERTIFICATIONS & LANGUAGES

Certification: Red Hat System Administration I (RH124)

Languages: French (fluent), English (fluent), Spanish (intermediate)